

Biometrik Case Study

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Biometrik Systems Pvt Ltd

Biometrik systems pvt Ltd is a technology company dealing with biometric identity management and AI security. It is based in Bangalore with its office in Mumbai. The company offers face recognition and face liveness detection services and products in the controlled industries: banking, financial services, and government. The Biometrik-developed technology solves one of the severe issues of the digital KYC systems since it allows not distinguishing between a real individual and an attempt to impersonate the system with the assistance of photos, videos, masks, or deepfakes. The solutions provide a secure customer onboarding process, video KYC, re-KYC, account access and fraud detection. They are compliant with the RBI, UIDAI and AML regulations.

Company Background & History

Mr. Ram Kunchur, the former CEO of DigiYatra, the Indian national face recognition project in the civil aviation industry, started Biometrik Systems Pvt Ltd between 2019 and 2021. Having managed a major aviation-based biometric implementation, Mr. Kunchur saw the necessity of systems of secure, spoof-proof, and scalable identity verification outside an airport, in particular, in the banking sector, telecommunications, government examination, and other controlled digital systems.

It began its operations as a deep-technology start-up, doing facial recognition algorithms, liveness systems that were patented, and QR-bound biometric identity solutions. This program was wholly built in India where both the software and hardware were produced locally.

Since its origin, Biometrik has developed to be an innovation-based startup through experience of rapid expansion, technology company, with enterprise, government and institutional clients, in India.

Stage of the Company

Biometrik Systems is at the stage of growth and expansion. The company is providing services to more than 15-20 enterprise and institutional customers in different sectors such as:

Aviation infrastructure (Adani Airports).

Government scrutinizing organizations (UPSC, civil services systems)

RTO and government projects.

Banking and financial services (Naika and several banks, conversations with HDFC Bank)

Telecom (including meetings with Reliance Jio)

Schools and institutions of education (Jharkhand and North India projects)

Digital identity in Himachal Pradesh at a regional level.

Target Market/ Industry Focus.

Biometrik Systems is mainly aimed at controlled and security sensitive industry where integrity of identity is very important. The main target markets are:

Banks of the Public and Privat Sector.

NBFCs and Digital Lending Platforms.

Telecom Operators

Aviation and Airport Authorities.

The government Examination Bodies.

Educational Institutions

Infrastructure Security and Enterprise Access Control.

Market Problem & Opportunity

Serious flaws in digital identity systems have been shown to be exposed by the swift digitization of financial and government services. The main issues of the present picture are:

Deep fake and artificial identity fraud.

Spoofing attacks on photos and video replay.

Creation of duplicate and mule accounts.

Liveness detection systems of low quality or those that can be overcome easily.

Threats regarding centralized identity storage.

Rising compliance and AML pressure.

Although there are numerous vendors of facial recognition tools, few have an integrated multi-layered authentication system using an integrated patent-backed system that provides:

Worldwide quality standards.

Liveness detection at an advanced level.

Biometric QR identity binding based on possession.

Decentralized identity structure.

On-premise and Flexible SaaS deployment.

Biometrik bridges this divide by providing a full MFA solution rather than a single biometric one.

Proprietary/ Patented Technology:

The anti-spoofing of authentication has been enhanced significantly by Biometrik which has come up with patented face liveness detection technology that has a near 100 percent accuracy in authentication in access control systems. It has a decentralized identity and self-sovereign identity (SSI) platform enabling users to securely and autonomously manage digital credentials. This will minimize the points of failure and enhance privacy. The solutions would be verifiable credentials, QR-based biometric identifiers, and the ability to integrate with existing hardware such as CCTV and readers to offer easy upgrade opportunities to enterprise and infrastructure security.

Key Strengths of Biometrik:

New core IP and patent on major biometric operations.

Multi-purpose deployment: on-premise, SaaS, or hybrid.

Advanced face reader, eGates and automated access hardware integrations.

Well-established leadership composed of a high degree of experience in biometric systems to face large-scale (e.g., DigiYatra, Cersai, UPSC Exam, to name a few).

Executive Summary

To protect the onboarding or verification of the customer, banks should have efficient multifactor authentication to make sure that the customer is real, alive and identifiable in addition to meeting the RBI KYC, AML, and audit requirements.

Biometrik Systems has bank-grade Multifactor Authentication (MFA) platform which includes:

Facial verification (1:1 and 1:N)

Liveness detection that is patented.

Optional secure QR-bound biometric credentials.

Audit-ready digital trails

The solution involves face recognition algorithm that has been evaluated independently and ranked as one of the best performers in the NIST Face Recognition Technology Evaluation (FRTE) guaranteeing the internationally accepted accuracy, fair and reliability. The platform is scalable, ready and API-driven.

to be integrated, so they smoothly fit into the current CBS, Video KYC systems of UCO Bank.

branch operations, mobile banking apps and mobile banking apps.

Goals of the Engagement.

Enforce powerful Multifactor Authentication as a means of verifying the customer identity.

Make sure that there is live customer presence in digital and video KYC.

Avoid impersonation, replaying attacks and synthetic identity fraud.

Biometric de-duplicate duplicate and mules.

KYC (key) Assistance, branch, and remote models.

Keep audit ready records of AML, compliance and regulatory review.

Facilitate an easy integration with the existing banking system.

Summary of Services and Capabilities.

The Biometrik solution is built as a Multifactor Authentication (MFA) platform of banking quality identity verification which integrates biometric identity, assurance of liveness, possession / context-driven factors and other attribute validation at all levels of the KYC process.

1.1 Face Match APIs - 1:1 and 1:N

Purpose: Customer onboarding, Video KYC, re-KYC, and biometric de-duplication of customers of UCO Bank.

Biometrik Systems offers APIs of Face Match that can be used to 1:1 verify as well as 1:N identify. They are driven by an algorithm that has been tested independently as part of the NIST Face Recognition Technology Evaluation (FRTE) and was ranked as one of the highest performing algorithms in the world.

1:1 Face Match is used to authenticate the face of a live customer with the reference image stored on KYC, CKYC, or bank records.

1:N Face Match is used to compare the face of a customer with large enrolled databases to detect duplicate, mule or even fraudulent identities.

Key Highlights:

Precision measured against NIST FRTE, which is a world-known evaluation program.

Written in large scale manner.

Applicable to mission-critical KYC, AML and fraud prevention operations.

1.2 Face Liveness Detection (Offline - Mandatory MFA Factor) competitive server-based (Offline - Mandatory MFA Factor)

Purpose: Verifies that the person being presented is a live individual, foiling spoofing attempts, such as deepfakes or masks, or photographs and videos. This constitutes an essential constituent of MFA in Digital and Video KYC.

On-Device (Offline) SDK works with the branch devices, field operation, and low connectivity areas.

Server Based SDK is designed to handle centralized Video KYC and high volume digital onboarding platforms.

Key Highlights:

Active and hybrid liveness detection which is patented.

Operates in other mode of lighting and devices.

Gain more trust at each verification point.

1.3 Face-in-QR code API.

Purpose: Binds the facial identity of a customer by a tamper-resistant QR code providing an additional authentication factor in assisted and branch-based KYC cases.

Creates patented Face-in-QR associated with customer identity.

Relates the facial biometrics to the customer reference credentials.

Key Highlights:

Eliminates QR cloning, reuse, and identity swapping.

Allows offline and assisted KYC check.

Applicable to onboarding branches, servicing accounts, and supported KYC.

1.4 Face Verification API (QR-Based).

Purpose: Ensures customer identity by comparing a live face with the face in the QR code and this forms a two-factor authentication (possession + biometric + liveness) process.

Conducts 1:1 verification and liveness verification.

Mobile/branch based applications, which can be used by bank employees.

Key Highlights:

Fast and operator-free MFA authentications.

Eliminates paperwork verifications.

Increases efficiency in the branch and customer satisfaction.

1.5 Age & Gender Detection

Purpose:

This feature provides a layer of validation that is not intrusive as it approximates age range and gender traits in live facial capture Digital KYC and Video KYC. It is a supplementary signal to enhance fraud detection, risk evaluation, and data conformity checks, which is not meant to be a major authentication determinant.

Real-time (not actual) age estimation.

Finds gender features of live face recognition.

Support background check as a silent run when doing KYC.

Outcomes:

Identifies deviations or discrepancies between reported KYC information and live capturing.

Enhances risk assessment of fraud when used along with MFA decision making.

Favors a privacy-first design and regulation-compliant KYC.

Face Authentication Each of the Lifecycle Stages of Faculty Attendances 1: One-Time Customer Registration / Onboarding.

Single face registration in Digital or Video KYC.

Mandatory liveness check

Mapping using Customer ID, Account or CKYC reference.

Result: Every customer receives a personal trusted digital identity.

Stage 2: Digital KYC / Re- KYC Authentication.

Face authentication through mobile application, web or branch system 1:1.

Each authentication attempt can be confirmed as lively.

Audit-logged and time labeled verification logs.

Deliverable: Customer authentication is authentic, it is regulated and it cannot be tampered with.

Stage 3: Faculty Records De-duplication.

One-to-many face similarity between the database.

Identifies duplicate, mule or suspicious identities.

Outcome: UCO Bank systems have one customer with one identity.

Stage 4: Compliance, Audit and system integration.

Shares verification is registered safely with AML, fraud detection and visibility.

Automatically creates audit trails and verification reports.

Result: Identity verification lifecycle has been prepared to undergo audit and complies.

Innovative Accuracy and Fairness Management.

Dynamic Threshold Management.

Adaptive thresholds trade accuracy and false matches.

Specialized Twin Handling

This was made to deal with genetically similar people with enhanced protection. These are measures that guarantee equity, inclusiveness, and sound operations.

Platform Compatibility & Integration.

Supported Operating Systems.

- o Windows
- o Linux
- o Android

Integration Characteristics

- o API-driven structure
- o Client and SDKs on the server-side.
- o Simple connection with CBS, Video KYC, AML, CKYC and digital banking systems.

Platform Compatibility Mobile App Mobile and Application Deployment:

KYC Customer and assisted through Android and iOS apps. The deployment of apps can be achieved via enterprise channels or app stores.

Supports:

- Face authentication
- Liveness detection
- MFA workflows
- Secure sync offline capture.

Web-based dashboards for:

- Branch officials
- Compliance and risk teams
- Monitoring central operations and units.

Competitive Landscape

Biometric identity and digital authentication is a market that consists of local and international vendors of facial recognition and KYC technology. Biometrik can be distinguished by:

- Liveness detection technology patented.
- Algorithm is among the 25 best in the world.
- Awareness at national-level hackathons.
- Complete make in India (hardware and software) development.
- Large-scale identity deployment experience.
- Biometric embedded identity innovation-QR.

Business Model

Biometrik has a loose Software-as-a-Service (SaaS) and Enterprise Licensing business model. Revenue streams include:

- Monthly subscription plans (pricing begins at approximately [?]49-[?]50/user in case of simple usage)

- Enterprise prices based on API pricing.
- Enterprise contracts that are based on volume.
- On-premise deployments licensing.
- Government and institutional deployments offerings.

Pricing varies based on:

- Complexity of use case
- Volume of transactions
- Environment (cloud, on-premise, or hybrid)

- Security requirements and compliance requirements.

This scalable pricing is one that is compatible with large enterprises as well as smaller organizations.

Risks & Challenges

Like any other sophisticated biometric technology vendor, some risks and issues exist:

- Blistering developments in deepfaking and AI-assisted spoofing.
- New laws with respect to privacy and protection of biometric data.
- Lack of acceptance in the semi-urban and rural settings.
- Threat of substitute identity solutions providers in the world.
- Continuous retraining of AI models and their performance.

Biometrik mitigates these risks by continuously developing and innovating its products, designing them with privacy in mind, being compliant with the laws, and maintaining a close working relationship with its institutional partners.

Strategic Vision/Growth Roadmap.

Biometrik Systems is an Indian-based company that focuses on becoming the leading provider of the biometric identity infrastructure and expanding worldwide. Strategic priorities are:

- On-going improvement of AI models to deal with new fraud attacks.
- Diversifying to global regulated markets.
- Forming more solid relationships with telecom and banks.
- Releasing performance validation and white papers.
- Developing rural and semi-urban deployable scalable solutions.
- Staying in line with international standards like GDPR and international compliance models.

The necessity of technological leadership and keeping up with changes in regulations, finances, and cybersecurity will make sustainable growth rely on changes in these areas.

Conclusion

Biometrik Systems provides an end-to-end and reputable Multifactor Authentication (MFA) system that enhances the digital KYC, re-KYC, and customer identity verification processes by UCO Bank without increasing operational complexity. The solution ensures trust, compliance, and continuity of customer identity supported by NIST FRTE-tested biometric accuracy, patented face liveness innovations, flexible deployment models, and trust.